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Examination	University	Institute	N Year	CPI/%
Graduation	IIT Bombay	IIT Bombay	2028	9.59/10
Intermediate	TSBIE	Sri Chaitanya Junior Kalasala	2024	97.2%
Matriculation	BSEAP	Sri Viswa Shanthi High School	2021	100%

SCHOLASTIC ACHIEVEMENTS

- Member of the **North Mumbai Zonal Toppers team** at **Mimamsa**, organised by **IISER Pune** (2026)
- Secured **5th rank** at **IITB** and **35th rank** globally in the **Simon Marais Mathematics Competition** (2025)
- Honored with the **Institute Academic Award** for outstanding performance among **1400+ students** (2025)
- Awarded prestigious **Reliance Foundation Merit Scholarship** for outstanding undergraduate studies (2024)
- Secured **All India Rank 10** in the **IIT Joint Entrance Examination Advanced** among 2 lakh candidates (2024)
- Achieved **State-wide Rank 12** in **AP EAMCET Exam 2024** out of **280,000+ candidates** state-wide (2024)

OLYMPIADS

- Among the top 35 students in India to clear the **Indian National Physics Olympiad (INPhO)** and invited **twice** to attend the **Orientation-Cum-Selection Camp for International Physics Olympiad (IPhO)** (2023,24)
- Awarded **Gold Medal** at **Orientation-Cum-Selection-Camp (OCSC)** for International Physics Olympiad (2023)
- Ranked among **Top 29** participants nationwide in **INAO** and was invited to **Astronomy OCSC** for **IOAA** (2024)
- Secured **All India Rank 4** in the **International Earth Science Olympiad (IESO)** in 8-th grade (2020)
- Awarded the prestigious **Kishore Vaigyanik Protsahan Yojana (KVPY)** Fellowship in 2021–2022 (2022)

RESEARCH INTERNSHIPS

Using Lean to Teach Proof Writing (SUAMI) | *Carnegie Mellon University* (May - July 2025)

Guide: Prof. Pavel Kovalev

- Developed an **interactive Lean 4 game** using the **Lean4 Game Engine**, modeled after the early chapters of **An Infinite Descent into Pure Mathematics-Clive Newstead**, allowing players to improve their **formal reasoning skills**
- Project serves as an optional study supplement and **under consideration by CMU** for integration into courses
- Implemented lightweight **custom tactics** in **Lean** emulating textbook strategies such as contradiction, contrapositive, introduction and elimination rules allowing beginners to **focus on ideas** rather than **Lean syntax** in the game

Exploring the Kinematics and Substructure of M31 & M32 (SIP) | *UCSC* (May - July 2024)

Guide: Prof. Raja GuhaThakurta, Rohit Raj

- Used data analysis libraries (**NumPy/Pandas/Astropy**) to process **SPLASH (Keck/DEIMOS)** and **DESI** survey data of **M31 (Andromeda)** and its satellite **M32**, designing methods to reduce noise caused by stellar clustering
- Evaluated **calibration accuracy** by comparing **DESI** data with the **HST-catalog** - the most precise in that region
- Built a comparison pipeline to conduct comparison of **SPLASH** and **DESI** measurements of velocity and intensity, aimed at understanding **M31 and M32** in the broader context of **investigating the kinematics and substructures**

PUBLICATIONS

Solving the N-Queens Problem Using Simulated Annealing | *[link]* (2024)

- Published research presenting a novel hybrid algorithm combining **simulated annealing** with **genetic operators** to solve the classical **N-Queens problem**, providing a unique optimization perspective in algorithmic development
- Demonstrated improved performance over classical backtracking by applying **probabilistic and evolutionary methods** to minimize conflicts efficiently while solving **N-Queens**, ensuring scalability and efficiency of the solutions
- Applied initialization, **iterative annealing** and **fitness-based selection** for scalable higher order **N-Queens**

QUANT EXPERIENCE

Quantitative Research Intern | *Quadeye* (May 2026 - July 2026)

- Exploring **ensemble pruning** and model compression for large-scale tree ensembles to **reduce inference time** without losing accuracy and f1, while building **reproducible pipelines** for pruning, validation and comparison
- Evaluated pruned ensemble models and neural networks for **correctness, runtime, scalability and generalizability**

Optiver Future Focus | *Mumbai* (Jun 2026)

- Studied **market making, options pricing, volatility, and algorithmic-trading** through lectures and workshops
- Applied concepts in **structured trading games, commodity mocks, and Optibook simulations** to analyze risk, execution, pricing inefficiencies, and strategy design under competitive conditions and **real market constraints**

Jane Street SEE | *Hong Kong* (Dec 2025)

- Studied **probability, market structure, and arbitrage** through lectures emphasizing quantitative reasoning
- Applied concepts in **mock trading games** based on real life market data to analyze risk, execution, pricing inefficiencies, and basic arbitrage opportunities under competitive conditions and **realistic market constraints**

KEY PROJECTS

Dynamic Routing Engine | Course Project: Data Structures and Algorithms Lab (November 2025)

Guide: Prof. Ashutosh Gupta

- Engineered a **graph routing system** using **K-D tree**, **A***, and **Yen's K-shortest path algorithms** to handle real-time navigation, traffic updates, and dynamic constraints across **100k+ node** with sub-second response times
- Developed intelligent delivery scheduling solutions for **NP-hard routing problems** using constraint-based heuristics, achieving **30% faster** delivery completion through optimal order batching and fleet management optimization
- Built C++ architecture with **JSON-based APIs** supporting **Euclidean/Manhattan/shortest-path metrics**, traffic-aware routing with **96-interval** speed profiles, and **10-20x faster** approximate queries with **95% accuracy**

Chess Engine with Classical Game Theory + NNUE | Seasons of Code (May - July 2025)

- Developed a chess engine in C++, performed **Min-max search with Alpha-Beta pruning** to increase search depth
- Trained a **NNUE (Efficiently Updatable Neural Network)** model on extensive board-value data from Stockfish
- Integrated **NNUE with iterative deepening search**, Zobrist hashing and transposition tables to increase efficiency and achieve rating of 2700, demonstrating strong competitive play and consistent performance against skilled players

Facial Recognition System | Winter in Data Science (December 2024)

- Built a **facial recognition system** with 97% accuracy using TensorFlow for deep learning and OpenCV for efficient real-time image capturing, GPU-accelerated pre-processing, optimized feature extraction, and latency reduction
- Performed **HOG-based detection**, extracted 128D embeddings and applied **distance-weighted k-NN**, combined with pre-processing (grayscale, resizing, normalization), PCA and evaluation on labeled datasets for accuracy

OTHER PROJECTS

xv6 Kernel Extensions | Course Project: OS, Guide: Prof. Purushottam (Jan 2026 - Apr 2026)

- Implemented **kernel-level multithreading** in xv6 using clone and join, enabling independently schedulable threads to share an address space and file descriptors while maintaining separate stacks, trapframes, execution state
- Implemented **futex-based synchronization** in xv6, including sleeplocks and semaphores, and applied them to a **producer-consumer** and **reader-writer** workloads using **sleep/wake** mechanisms to eliminate busy waiting
- Enhanced the xv6 scheduler with **priority scheduling**, **CPU affinity**, and **context-switch tracking**

Distributed Algorithms Simulation Project | IIT Indore, Prof. Debasish (Dec 2025)

- Extended an existing **distributed-algorithm simulator** (based on the CCM Model) by implementing a new algorithm for **coordinated distribution of k agents across a graph**, where agents operate independently
- Implemented multiple strategies for agent dispersion, compared their **efficiency** through large scale experiments
- Built **modular simulation components** enabling rapid prototyping of new agent distribution algorithms

Circuit Synthesis Using SAT Solver | Course Project: Logic, Guide: Prof. Krishna S (August 2025)

- Built a SAT-solver based framework to automatically **synthesize NAND-only circuits** from arbitrary Boolean formulae/ truth tables while **minimizing the number of gates** through iterative deepening on number of gates
- Implemented in **Python** with PySAT, encoded input formulas into truth tables and applied NAND logic to generate valid NAND connections with constraints for input encoding, gate count, logic semantics and output constraints

Assorted Projects | Self Projects (2023-Present)

- Built a two-player **Angry Birds-style game** in **pygame-ce** with real physics, wind, themes and custom power-ups
- Built a SAT solver in c++ using DPLL/CDCL with clause learning, intelligent backtracking, and efficient heuristics
- Built a **number recognition system** entirely in **NumPy**, implementing forward/backward propagation and the **Adam optimizer**, without any external ML libraries to train and evaluate a small neural network from scratch
- Developed a **Rubik's Cube solver** in Python using the IDA* algorithm with a 54-element state, misplacement-based heuristic, move-cycle tables, pruning of inverse/duplicate states and depth-bounded search for efficient solutions
- Implemented a **Sudoku solver** using recursive backtracking with constraint checks and logging for real-time steps
- Designed a **spell-checker** with a Trie-based dictionary, enhanced **Levenshtein distance** using keyboard proximity and frequency-weighted ranking for accurate, context-aware auto-correction and word auto-complete suggestions
- Built a **MCMC** tool for decrypting basic substitution encryption and evaluating performance on noisy text data

TECHNICAL SKILLS

Languages	Python, Bash, HTML, CSS, JavaScript, Make, Matlab, Assembly, Verilog, MIPS
Software/Tools	Git, GitHub, Jupyter, L ^A T _E X, Sed, Awk, Fusion360, Flask, FastAPI, SQLite, Astropy
Data Science	NumPy, Pandas, Matplotlib, TensorFlow, Keras, PyTorch, OpenCV, Scikit-learn

EXTRACURRICULAR ACTIVITIES

- Selected for Phase 2 of **Graviton Crucible**, learning market making through trading games and lectures (2026)
- Participated in **ACM Summer School 2026** at IIT Bombay on Foundations of **Hardware Security** (Jun 2026)
- Designed and built a **quadcopter** from the ground up, including a custom joystick PCB using Arduino, frame modeling with Fusion 360, laser-cut fabrication and integrated a PID controller to achieve stable control (2025)
- Participated in the **NSO athletics** training program, developing discipline, endurance and teamwork skills (2025)
- Awarded full scholarship by **Sri Chaitanya School**, with stipend support to join advanced training (2024)
- Achieved **third place** in **Tyro CTF Competition** across all undergraduates (IIT Bombay) (2024)
- Trained under-resourced students** in Mathematics and Science for 2 years, improving local literacy (2020-2024)
- Taught Yoga and Meditation to local community kids, promoting health, mindfulness and awareness (2020-2024)